

Deep Learning-Assisted Initial Iteration Values Selection for Fracture Propagation Simulation using Displacement Discontinuity Method

Purpose

The purpose of this paper is to address the high sensitivity and convergence instability of Newton-Raphson (NR) solvers to initial iteration values in hydraulic fracture propagation simulations.

Design/methodology/approach

The paper proposes a novel initialization method utilizing physics-informed neural networks (PINNs). The governing equations for fracture propagation, derived via the displacement discontinuity method (DDM), are directly embedded into the neural network's loss function. The network is subsequently trained using a hybrid strategy of Adam and L-BFGS optimizers to ensure physical consistency without relying on labeled data.

Findings

The proposed method significantly enhances solver stability and computational efficiency. Experimental results demonstrate that the proposed method decreases the divergence rate associated with random initialization by 27.22% and reduces the average number of iterations from 16.79 to 2.0.

Originality/value

This paper provides a highly reliable initial value selection approach that resolves the persistent computational bottleneck of NR solvers in DDM-based fracture simulations. Furthermore, it establishes a transferable physics-informed deep learning paradigm for improving the robustness of solvers in other nonlinear multi-physics systems.

Keywords:

Hydraulic fracturing, Physics-informed neural networks, Displacement discontinuity method, Newton-Raphson solver, Initialization method

1 Introduction

Hydraulic fracturing is a crucial technology for enhancing hydrocarbon recovery.

Perkins and Kern and Nordgren first proposed the PKN model to describe the evolution of fracture length and width during hydraulic fracturing (Perkins and Kern 1961;

Nordgren 1972). Geertsma and de Klerk developed the KGD model, which is commonly used to characterize hydraulic fracture growth at the early stages of fluid injection (Geertsma and De Klerk 1969). These two classical models established the theoretical basis for the simulation of hydraulic fracturing. Since then, many researchers have begun to focus on the mathematical modeling of fracture propagation. Simonson et al. (1978) studied the relationship between in-situ stress and fracture height growth, contributing to the development of pseudo-three-dimensional (P3D) models. Clifton and Abou-Sayed (1979) proposed the planar three-dimensional (PL3D) model to incorporate the non-local elastic response, and solved full 3D elasticity equations using a moving mesh system.

In recent decades, numerical methods have been applied to simulate hydraulic fracturing, such as the Finite Element Method (FEM), the Discrete Element Method (DEM) (Ismail and Azadbakht 2024; Wang et al. 2025). Chen et al. (2016) simulated the interaction between hydraulic and natural fractures by using FEM with a cohesive zone model. Dong et al. (2021) modeled the propagation of complex hydraulic fracture networks in naturally fractured reservoirs using the Extended Finite Element Method (XFEM). Tang et al. (2022) analyzed fracture propagation mechanisms in glutenite reservoirs by adopting the DEM. In contrast to these domain-based methods, the Displacement Discontinuity Method (DDM) significantly reduces the problem dimension and offers superior computational efficiency by discretizing only the fracture boundaries rather than the entire rock volume (Crouch S. L. 1976). Wu and Olson (2014) modeled the simultaneous propagation of multiple hydraulic fractures by employing DDM to capture the stress interference between closely spaced fractures. Li et al. (2025) investigated asymmetric fracture initiation using DDM, finding that

fractures gradually reorient under local stress fields to form approximately symmetric trajectories away from the wellbore.

Given the strong nonlinearity and multi-physics coupling involved in hydraulic fracturing (Detournay 2016; Lecampion et al. 2018), the Newton–Raphson (NR) method has been widely adopted to solve the resulting system of nonlinear equations (Wu 2014; Wu and Olson 2014; Zheng 2021; Zhang et al. 2024). Since the NR method is a locally convergent algorithm that is highly sensitive to initial values, a continuation strategy is frequently employed to ensure numerical stability in DDM-based simulations of fracture propagation. The converged solution from the previous time step is used as the initial values for the NR iteration at the current step, which effectively reduces the number of iterations and accelerates convergence (Wu 2014; Yang et al. 2022).

However, this strategy cannot be applied to the initial stage of fracture propagation, because no prior solution is available to serve as the initial values. An inappropriate initialization may lead to numerical divergence or a dramatic increase in the number of iterations, which severely limits computational efficiency in practical engineering applications. To address this, this paper proposes a deep learning-based initialization method to provide accurate initial values, thereby overcoming these instabilities and ensuring reliable simulation performance.

Deep learning has been widely adopted in the field of scientific computing due to its strong capability to approximate highly nonlinear relationships (Yue et al. 2023; Zhao et al. 2024; Liu et al. 2025). Raissi et al. (2019) developed Physics-Informed Neural Networks (PINNs), which incorporate physical priors directly into the loss function to enforce physical constraints and significantly reduce data dependency. Recently, PINNs have been successfully applied to the field of hydraulic fracturing. Lu et al. (2022) integrated the PKN model with PINNs to achieve accurate fracture

geometry prediction. Pu et al. (2022) employed PINNs for reservoir flow field modeling and demonstrated that physical constraints enable accurate predictions even with scarce training samples. Ryu et al. (2025) extended the architecture to a dual-network system to satisfy the constraints of moving boundary during the spatiotemporal evolution of fracture propagation. Inspired by the PINN framework, this paper proposes a novel method to accurately select initial values for the NR solver in hydraulic fracturing simulations. The proposed method embeds the governing equations of fracture propagation formulated by DDM into the loss function, ensuring satisfaction of physical constraints. Numerical experiments demonstrate that the proposed method significantly improves the convergence robustness and computational efficiency of the NR solver.

The remainder of this paper is organized as follows: In Section 2 we establish the fracture propagation model based on DDM. Section 3 details the proposed physics-informed deep learning framework, followed by Section 4 which presents the experimental results. Finally, Section 5 concludes this work.

2 Fracture Propagation Model

Given that initially asymmetric fractures evolve toward approximately symmetric geometries during propagation (Li et al. 2025), we just consider symmetric fractures. Suppose the vertical fracture has constant height and propagates within a homogeneous, linear elastic, and isothermal reservoir. Only Mode I (opening) and Mode II (shearing) deformations are considered (Figure 1), with planar Poiseuille fluid flow assumed and proppant transport neglected.

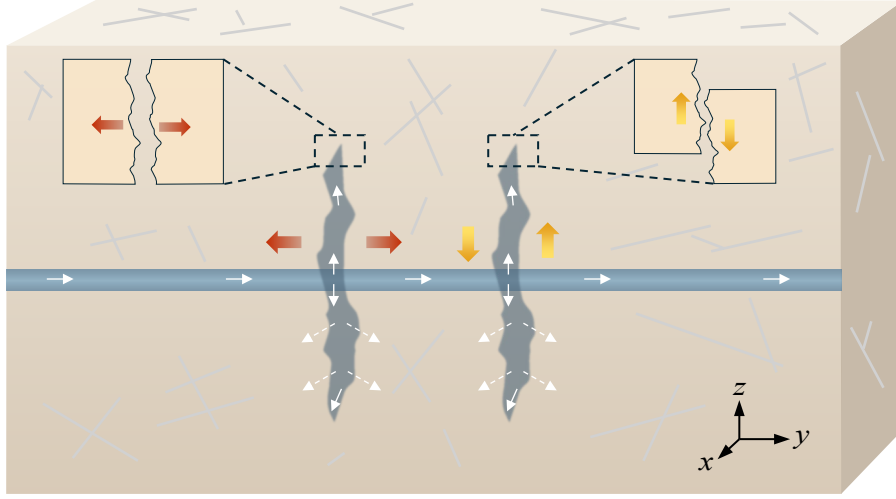


Figure 1. Schematic of fracture modes: Mode I (opening) and Mode II (shearing).
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The DDM is employed to simulate non-planar hydraulic fracture propagation. Figure 2 illustrates the discretization of a fracture in the DDM. The fracture is divided into N segments along its length. A global coordinate system (X, Y) and a local coordinate system (x, y) for each segment are defined. In the local system, the x -axis is tangential to the segment, while the y -axis is normal to the segment surface. Based on the above discretization, this paper develops a coupled model involving rock deformation, fluid flow in the fracture and fracture propagation.

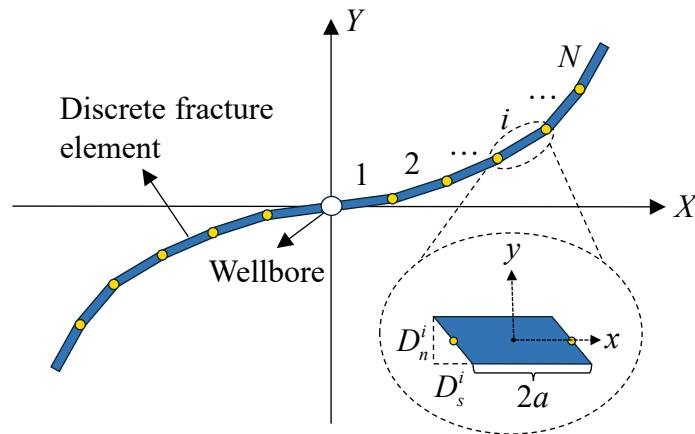


Figure 2. Schematic of the discretization of a symmetric fracture.
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In an elastic domain, the stress changes induced by fractures can be computed using the principle of linear elastic superposition. On this basis, the total normal and shear stress changes on the i -th segment are obtained by summing the contributions from all fracture segments within the domain (Olson 2004).

Due to the geometric symmetry of the fracture with respect to the wellbore, only the N segments along a single wing are considered. The contribution of the symmetric fracture segment $\bar{j}$ to segment i is incorporated through an effective influence coefficient $K_{\alpha\beta}^{ij}$:

$$K_{\alpha\beta}^{ij} = G^{ij} C_{\alpha\beta}^{ij} + G^{\bar{i}\bar{j}} C_{\alpha\beta}^{\bar{i}\bar{j}}, \quad (i, j = 1, \dots, N; \alpha, \beta \in \{n, s\}) \quad (1)$$

where G^{ij} and $G^{\bar{i}\bar{j}}$ represent the three-dimensional correction factors, and $C_{\alpha\beta}^{ij}$ and $C_{\alpha\beta}^{\bar{i}\bar{j}}$ are the stress influence coefficients of segment j and its symmetric counterpart $\bar{j}$ on segment i , respectively.

During the fracture propagation process, each fracture segment satisfies the stress equilibrium condition, which can be expressed as (Wu 2014):

$$\begin{cases} \sum_{j=1}^N (K_{ns}^{ij} D_s^j + K_{nn}^{ij} D_n^j) = p_i - \sigma_h \cos^2 \theta_i - \sigma_H \sin^2 \theta_i, \\ \sum_{j=1}^N (K_{ss}^{ij} D_s^j + K_{sn}^{ij} D_n^j) = \sigma_s^{\text{local}} - \frac{1}{2}(\sigma_h - \sigma_H) \sin 2\theta_i, \end{cases} \quad (i = 1, \dots, N). \quad (2)$$

where N denotes the total number of segments along a single wing of the fracture, D_n^j and D_s^j represent the normal and shear displacement discontinuities of the j -th segment, p_i is the fluid pressure within the i -th segment, θ_i is the inclination angle of the i -th segment starting anticlockwise from the X -axis, σ_H and σ_h are the maximum

and minimum principal stresses, respectively, and σ_s^{local} is the local shear stress, which becomes zero when the fracture opens. See Appendix A for the detailed derivation.

The fracturing fluid is assumed to be non-Newtonian and incompressible, with the flow modeled as plane Poiseuille flow. Based on the Navier-Stokes equations, the relationship between pressure gradient and flow rate is governed by (Valkó and Economides 1995):

$$\frac{\partial p}{\partial s} = \frac{1}{\beta_1 \beta_2} 2^{\eta_1+1} \eta_2 \left(\frac{1+2\eta_1}{\eta_1} \right)^{\eta_1} h^{-\eta_1} w^{-(2\eta_1+1)} q^{\eta_1}, \quad (3)$$

where s is the distance along the fractures from the wellbore, p is fluid pressure, q is flow rate, w is fracture width, h is fracture height, η_1 is the fluid power-law index, η_2 is the consistency index, and β_1, β_2 are unit-conversion factors.

Considering the effect of fluid leak-off, the injected fluid satisfies the mass conservation equation (Wu 2014):

$$\frac{\partial q(s,t)}{\partial s} - \frac{2hC_L}{\sqrt{t-\tau(s)}} - \frac{\partial A(s,t)}{\partial t} = 0, \quad (4)$$

where C_L is the leak-off coefficient, $A(s,t)$ is the cross-sectional area, and $\tau(s)$ is the arrival time of the fracture tip at position s .

Hydraulic fracture propagation behavior is determined by the stress state at the fracture tip and the mechanical properties of the rock. During fracture propagation, the normal stress at the fracture tip is zero (Wu 2014):

$$p_{\text{tip}} - \sigma_H \sin^2 \theta_{\text{tip}} - \sigma_h \cos^2 \theta_{\text{tip}} = 0. \quad (5)$$

Here, the tip fluid pressure p_{tip} is obtained through linear extrapolation from adjacent segments:

$$\begin{cases} p_{\text{tip}} \approx \frac{3}{2} p_N - \frac{1}{2} p_{N-1}, & N > 1, \\ p_{\text{tip}} \approx 2p_1 - p_w, & N = 1. \end{cases} \quad (6)$$

where p_N and p_{N-1} are the pressures at the centers of the N -th and $(N-1)$ -th segments, respectively, and p_w is the bottom-hole pressure.

With the fracture geometry and pressure distribution determined, the Mode I (opening) and Mode II (shearing) stress intensity factors K_I and K_{II} are calculated as (Olson 2008):

$$\begin{cases} K_I = \frac{0.806E\sqrt{\pi}}{4(1-\nu^2)\sqrt{\Delta x}} D_n, \\ K_{II} = \frac{0.806E\sqrt{\pi}}{4(1-\nu^2)\sqrt{\Delta x}} D_s. \end{cases} \quad (7)$$

where E is Young's modulus and ν is Poisson's ratio.

Based on the maximum circumferential stress criterion, fracture propagation occurs when the opening-mode stress intensity factor exceeds the rock fracture toughness. The propagation direction is shown in Figure 3 and can be expressed as (Erdogan and Sih 1963):

$$\begin{cases} \tan \frac{\Delta\theta}{2} = \frac{1}{4} \frac{K_I}{K_{II}} + \frac{1}{4} \sqrt{\left(\frac{K_I}{K_{II}}\right)^2 + 8}, & K_{II} < 0, \\ \tan \frac{\Delta\theta}{2} = \frac{1}{4} \frac{K_I}{K_{II}} - \frac{1}{4} \sqrt{\left(\frac{K_I}{K_{II}}\right)^2 + 8}, & K_{II} > 0. \end{cases} \quad (8)$$

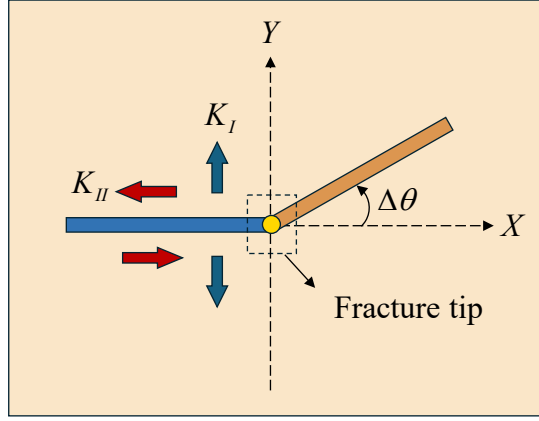


Figure 3. Schematic of fracture propagation direction.

Source: Authors own work

In summary, Equations (2),(3),(4) and (5) form a fully coupled system. For the initial fracture segment, this system is solved simultaneously to determine the five unknown variables: D_n^I , D_s^I , p_1 , p_w , and Δt . During subsequent propagation, the converged values from the previous segment are used to initialize the current segment, with the propagation direction determined by Eq. (8).

3 Methodology

The proposed PINN framework and experimental setup are detailed in this section.

3.1 PINN Architecture

As defined in Section 2, the target physical variables are denoted as the vector given by $\mathbf{u} = [D_n, D_s, p_f, p_w, \Delta t]^T$. To predict the solution variables under different parameter conditions, a PINN framework is developed to establish the mapping from the initial fracture inclination angle θ and the injection rate q to $\mathbf{u}$. Specifically, the network learns a nonlinear mapping of the form:

$$\mathbf{u} = f(\mathbf{x}; \boldsymbol{\Theta}), \quad (9)$$

Where $\mathbf{x} = [\theta, q]^T$, f represents the DNN, and $\boldsymbol{\theta}$ are the parameters of the neural network.

Figure 4 shows the proposed architecture, which integrates a DNN with physical constraints. The input parameters $\mathbf{x}$ are propagated through multiple fully connected hidden layers with nonlinear activation functions to predict the solution variables $\mathbf{u}$. To improve training efficiency and numerical stability, sigmoid-based scaling is applied at the output layer to restrict the predictions to physically admissible ranges. The physical constraints are then formulated as residuals into a unified loss function, which is minimized to update the network parameters during training. This framework enforces multi-physics consistency without labeled data, thereby providing robust and physically consistent initializations for numerical simulations.

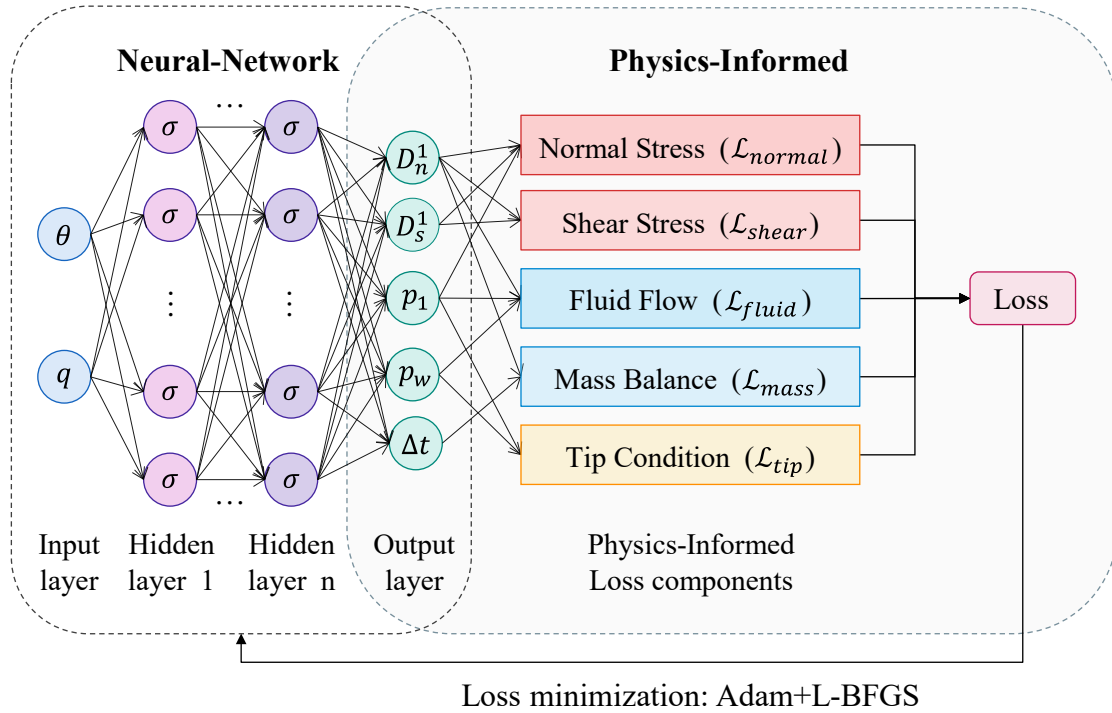


Figure 4. Architecture of the proposed model.

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3.2 Physics-Informed Training

The training of the neural network f is realized by minimizing a physics-informed loss function, which constrains the network parameters $\boldsymbol{\Theta}$ to satisfy the governing equations derived in Section 2. By embedding the physical constraints defined by Eqs. (2), (B-2), (B-7) and (5) and formulating their residuals as Mean Squared Error (MSE) terms, the total loss function is defined as:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\Theta}) = \mathcal{L}_{\text{normal}}(\boldsymbol{\Theta}) + \mathcal{L}_{\text{shear}}(\boldsymbol{\Theta}) + \mathcal{L}_{\text{fluid}}(\boldsymbol{\Theta}) + \mathcal{L}_{\text{mass}}(\boldsymbol{\Theta}) + \mathcal{L}_{\text{tip}}(\boldsymbol{\Theta}), \quad (10)$$

where each term is defined as follows:

$$\mathcal{L}_{\text{normal}}(\boldsymbol{\Theta}) = \frac{1}{N_c} \sum_{k=1}^{N_c} \left| K_{ns} D_s^{1,k} + K_{nn} D_n^{1,k} - (p_1^k - \sigma_h \cos^2 \theta_1^k - \sigma_H \sin^2 \theta_1^k) \right|^2, \quad (11)$$

$$\mathcal{L}_{\text{shear}}(\boldsymbol{\Theta}) = \frac{1}{N_c} \sum_{k=1}^{N_c} \left| K_{ss} D_s^{1,k} + K_{sn} D_n^{1,k} - \left(\sigma_s^{\text{local}} - \frac{1}{2}(\sigma_h - \sigma_H) \sin 2\theta_1^k \right) \right|^2, \quad (12)$$

$$\mathcal{L}_{\text{fluid}}(\boldsymbol{\Theta}) = \frac{1}{N_c} \sum_{k=1}^{N_c} \left| \frac{p_w^k - p_1^k}{a} - \frac{12\mu q_0^k}{\beta_1 \beta_2 h (w_0^k)^3} \right|^2, \quad (13)$$

$$\mathcal{L}_{\text{mass}}(\boldsymbol{\Theta}) = \frac{1}{N_c} \sum_{k=1}^{N_c} \left| -q_0^k \Delta t^k + 8haC_L \sqrt{\Delta t^k} + 2ahD_n^{1,k} \right|^2, \quad (14)$$

$$\mathcal{L}_{\text{tip}}(\boldsymbol{\Theta}) = \frac{1}{N_c} \sum_{k=1}^{N_c} \left| 2p_1^k - p_w^k - \sigma_h \sin^2 \theta_1^k - \sigma_H \cos^2 \theta_1^k \right|^2. \quad (15)$$

where $D_n^{1,k}$ and $D_s^{1,k}$ are the predicted normal and tangential displacement

discontinuities, p_1^k and p_w^k represent the fluid pressures at the fracture center and the

wellbore, and Δt^k indicates the propagation time step, N_c denotes the total number of collocation points.

To minimize the loss defined in Eq. (10), network parameters are initialized via the Xavier Normal scheme and updated using a hybrid strategy: the Adam optimizer is first employed to rapidly navigate to the vicinity of the optimal solution, followed by a transition to the second-order L-BFGS algorithm to enhance local convergence and numerical precision (Raissi et al. 2019). This detailed procedure is summarized in Algorithm 1, while the training loss curve in Appendix C further validates its effectiveness. This approach ensures strict adherence to the coupled physical constraints, achieving robust generalization without reliance on labeled data.

Algorithm 1. Training process for PINN.

Input: Training domain Ω , max iterations N_{Adam}, N_{L-BFGS}
Output: Optimized network parameters Θ^*

- 1: Sample collocation points $\mathcal{D}_c = \{(\theta_1^k, q_0^k)\}_{k=1}^{N_c}$ using LHS on Ω
- 2: Construct DNN $f(\mathbf{x}; \Theta)$ and initialize Θ with Xavier Normal scheme
- 3: $iter \leftarrow 0$
- 4: **While** $iter < N_{Adam} + N_{L-BFGS}$ **do**
- 5: Forward Pass: Compute $\hat{\mathbf{u}} \leftarrow f(\mathcal{D}_{col}; \Theta)$
- 6: Compute Loss: Calculate total loss $\mathcal{L}_{total}$ according to Eq. (10)
- 7: Backpropagation: Compute gradients $\nabla_{\Theta} \mathcal{L}_{total}$
- 8: **if** $iter < N_{Adam}$ **then**
- 9: Update Θ using Adam optimizer
- 10: **else**
- 11: Update Θ using L-BFGS optimizer
- 12: **end if**
- 13: $iter \leftarrow iter + 1$
- 14: **end while**
- 15: **return** Model parameters Θ^*

3.3 Experimental Setup

Given the symmetry of the in-situ stress field, restricting the inclination angle to $[0^\circ, 90^\circ]$ is sufficient to cover all inclination scenarios. Together with an injection rate ranging from 1 to $10 \text{ m}^3 / \text{min}$, the resulting parameter space is defined as

$\Omega = [0^\circ, 90^\circ] \times [1, 10] \text{ m}^3 / \text{min}$. Collocation points ($N_c = 10,000$) are then generated over

Ω using Latin Hypercube Sampling (LHS). The proposed PINN is implemented as a fully connected neural network with six hidden layers of 128 neurons each, employing the Tanh activation function in the hidden layers and a linear activation in the output layer. The output variables are mapped to the physical intervals: $D_s \in [-0.1, 0]$ m, $D_n \in [0, 0.1]$ m, $\Delta t \in [0, 0.1]$ d, and $P_f, P_w \in [0, 50]$ MPa.

Numerical experiments are conducted using Python 3.11.8 and PyTorch 2.2.2. The governing physical parameters are set as $\eta_1 = 1$, $\eta_2 = 100 \text{ mpa} \cdot \text{s}^n$, $\nu = 0.2$, $E = 45023 \text{ MPa}$, $h = 60.96 \text{ m}$, $\sigma_H = 31.37 \text{ MPa}$, $\sigma_h = 30.68 \text{ MPa}$, and $C_L = 3.05 \times 10^{-5} \text{ m/min}^{0.5}$. Finally, the model performance is evaluated against a NR iterative solver.

To evaluate the model performance, we employ the Absolute Error, MSE, and Relative L_2 Error to compare the predicted solution u_{pred} against the reference solution u_{ref} . The Absolute Error is defined as

$$E_{\text{abs}} = |u_{\text{pred}} - u_{\text{ref}}|, \quad (16)$$

which characterizes the point-wise local deviation between the predicted and reference solutions.

For quantifying the averaged prediction accuracy over the computational domain, the MSE is formulated as

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \left(u_{\text{pred}}^{(i)} - u_{\text{ref}}^{(i)} \right)^2, \quad (17)$$

where N denotes the number of sample points.

In addition, the Relative L_2 Error is introduced to assess the global discrepancy while eliminating dimensional effects, and is given by

$$\text{RE}_{L_2} = \frac{\|\mathbf{u}_{\text{pred}} - \mathbf{u}_{\text{ref}}\|_2}{\|\mathbf{u}_{\text{ref}}\|_2} = \sqrt{\frac{\sum_{i=1}^N (u_{\text{pred}}^{(i)} - u_{\text{ref}}^{(i)})^2}{\sum_{i=1}^N (u_{\text{ref}}^{(i)})^2}}. \quad (18)$$

4 Results and Discussion

This section describes the performance of the proposed model, including accuracy validation against numerical benchmarks, stability and efficiency analysis, sensitivity analysis of the network architecture, and application to fracture propagation simulations.

4.1 Result Analysis

The accuracy of the proposed model is evaluated by comparing its predictions with numerical benchmarks over the sampled parameter space. As shown in Figure 5, the predicted fields exhibit close agreement with the reference solutions across all physical variables under varying parameter combinations. The absolute errors for D_n^1 , p_1 , and p_w are more than three orders of magnitude lower than their reference values. For D_s^1 , the theoretical shear stress vanishes at $\theta = 0^\circ$ and $\theta = 90^\circ$. While this zero-value characteristic poses a challenge for neural network approximation, the prediction error remains at a low level. Finally, for Δt , despite the significant variations in the reference solution at low injection rates leading to relatively larger deviations, the absolute error is controlled within one order of magnitude.

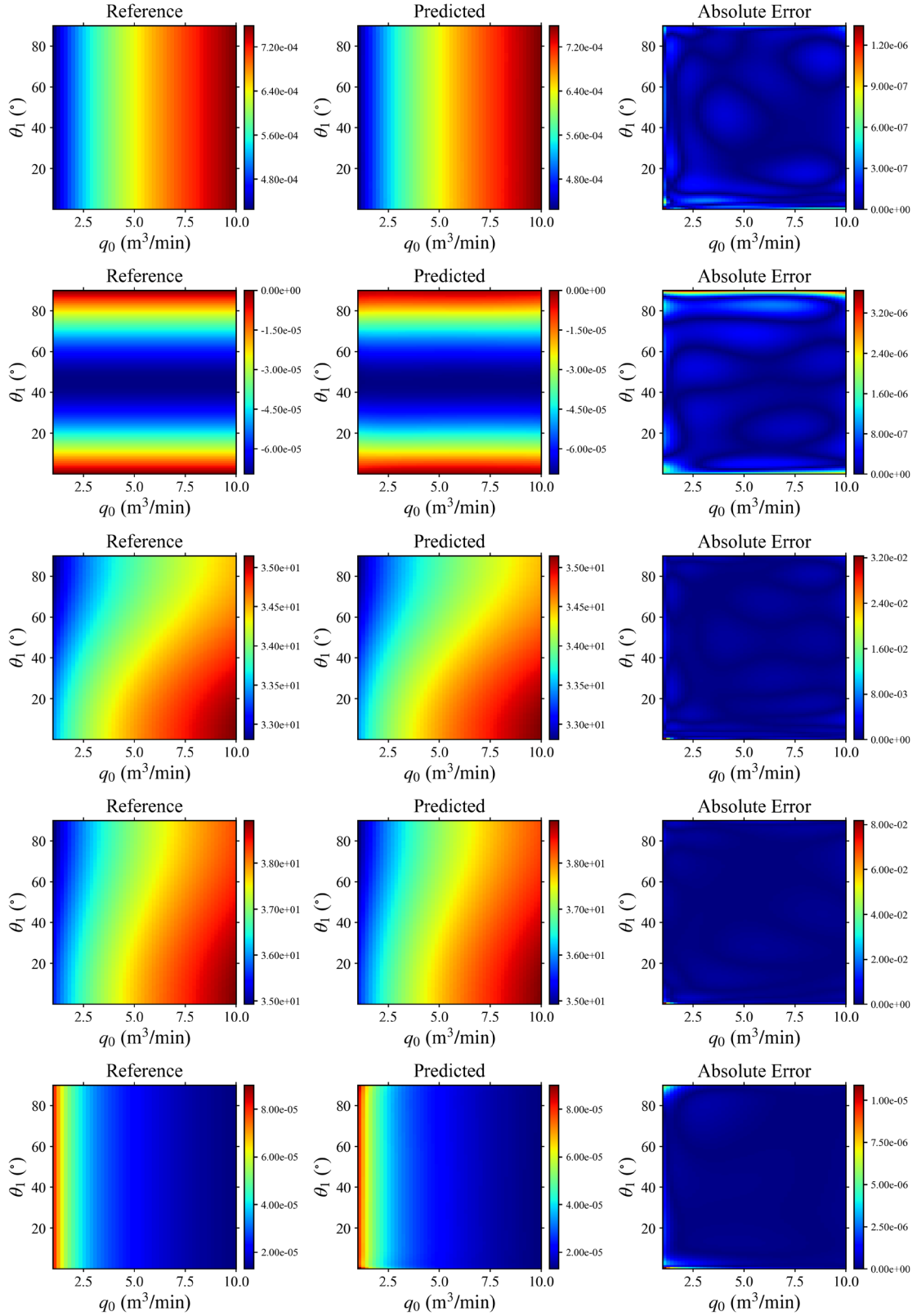


Figure 5. In order from top to bottom: $D_n^1, D_s^1, p_1, p_w, \Delta t$. Left: Numerical benchmark solution; Center: PINN predicted solution; Right: Absolute error distribution.
Source: Authors own work

To further quantify the model performance, Figure 6 presents a statistical evaluation of the prediction accuracy. As shown, the relative L_2 errors for all variables remain below 2×10^{-2} , with those for D_n , p_1 , and p_w below 1.5×10^{-4} . In addition, the MSE for all variables converges to the order of 10^{-6} or lower. Collectively, these results demonstrate that the model achieves high-fidelity predictions for all physical variables.

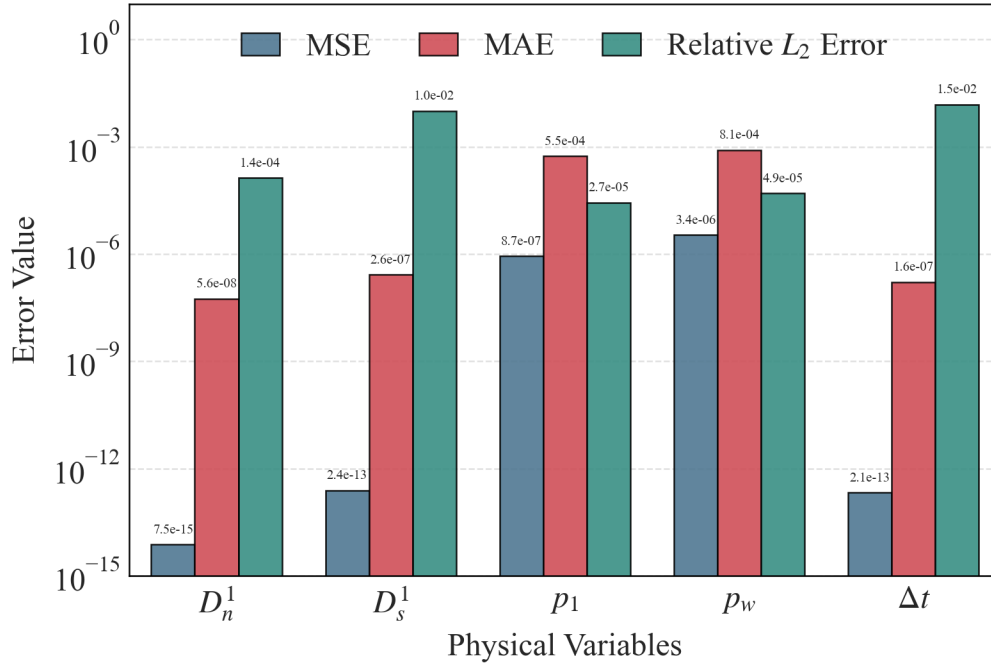


Figure 6. Error analysis of the predicted physical variables.

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4.2 Stability and Efficiency

The random initialization strategy within a given interval was adopted as a comparative baseline. As illustrated in Figure 7, while the solver converges in certain samples, it requires an average of 16.79 iterations. More critically, Figure 7(b) reveals that the method suffers from a high average divergence rate of 27.22%, indicating the instability of this initialization strategy.

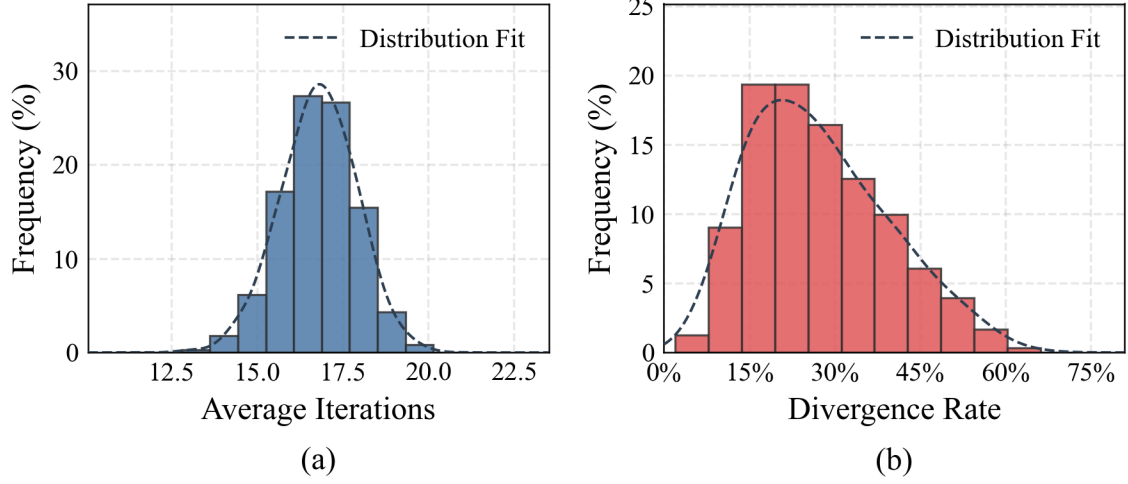


Figure 7. Statistical histograms of convergence performance for random initialization (based on 5,000 sample points in Ω with 50 Monte Carlo simulations per sample, where initial values are randomly sampled from $D_n^l \in [10^{-5}, 10^{-2}]$ m, $D_s^l \in [-10^{-2}, -10^{-5}]$ m, $p_1, p_w \in [30, 40]$ MPa, and $\Delta t \in [10^{-6}, 10^{-4}]$ d). (a) Distribution of average iterations for converged samples; (b) Distribution of average divergence rates.

Source: Authors own work

In contrast, the proposed model exhibits superior stability and computational efficiency. Figure 8 shows the performance metrics across varying sample sizes, indicating that the average iteration count is reduced to a stable value of 2.0, the computation achieves an approximate six-fold acceleration, and the divergence rate remains at 0% across all test cases. The fact that only two iterations are required indicates that the initial values generated by the proposed model already fall within the convergence basin of the NR method, which prevents numerical divergence, while the second iteration guarantees the high precision of the final solution. These results confirm that the proposed approach resolves the persistent bottleneck of initialization sensitivity in NR iterations, thereby improving the overall simulation robustness.

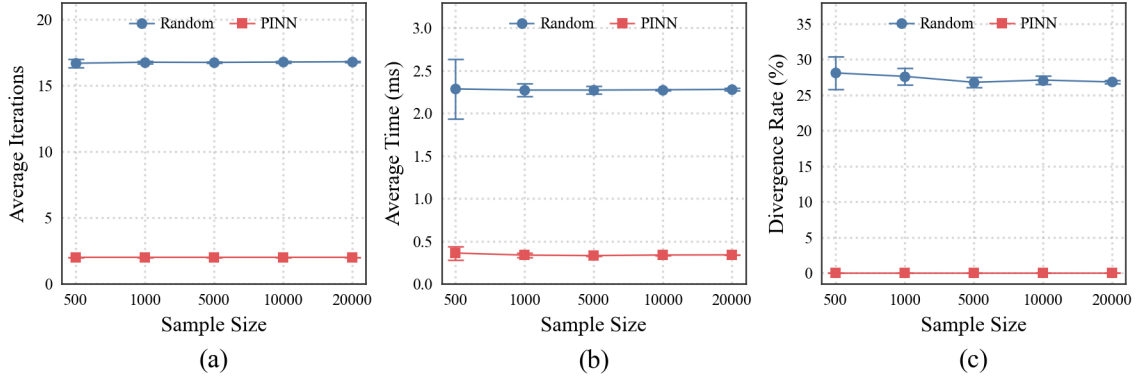


Figure 8. Performance comparison between random initialization and the proposed model-based initialization under different sample sizes. (a) Average iterations; (b) Average computation time; (c) Algorithm divergence rate. Data points represent the mean of 5 independent experiments; error bars indicate standard deviation.

Source: Authors own work

4.3 Sensitivity Analysis

To evaluate the sensitivity of the proposed model to hyperparameter configurations, we examined total loss variations across different network architectures. As shown in Table 1, shallow and narrow architectures lack sufficient expressive capacity, resulting in relatively high training loss. Although the approximation capability improves significantly as network depth and width increase, performance gains tend to saturate or even degrade at excessive levels of complexity. For example, with the width fixed at 128 neurons, the loss increased from 5.66×10^{-6} at 6 layers to 1.81×10^{-5} at 8 layers. Consequently, the architecture with 6 hidden layers and 128 neurons per layer achieved the lowest total loss of 5.66×10^{-6} and was selected as the optimal configuration.

Table 1. Comparison of the total loss under different combinations of hidden layers and neurons per layer.

Hidden Layers	Neurons per Layer			
	32	64	128	256
2	1.21×10^{-4}	2.03×10^{-5}	5.21×10^{-5}	2.16×10^{-5}
4	2.97×10^{-5}	8.14×10^{-6}	9.33×10^{-6}	6.61×10^{-6}
6	5.89×10^{-5}	3.16×10^{-5}	5.66×10^{-6}	6.88×10^{-6}
8	4.05×10^{-5}	2.45×10^{-5}	1.81×10^{-5}	8.28×10^{-6}

4.4 Fracture Propagation Simulation

Based on the proposed model, reliable initial values can be selected for the first fracture segment. Subsequently, the converged solution of the preceding step is utilized as the initial guess for the current step. As shown in Figure 9, the simulation effectively captures fracture reorientation, and the induced maximum principal stress fields display the typical wing-shaped pattern with pronounced concentration at the tip, thereby validating the physical rationality of the simulation. This recursive initialization strategy establishes a continuous, stable, and highly efficient workflow.

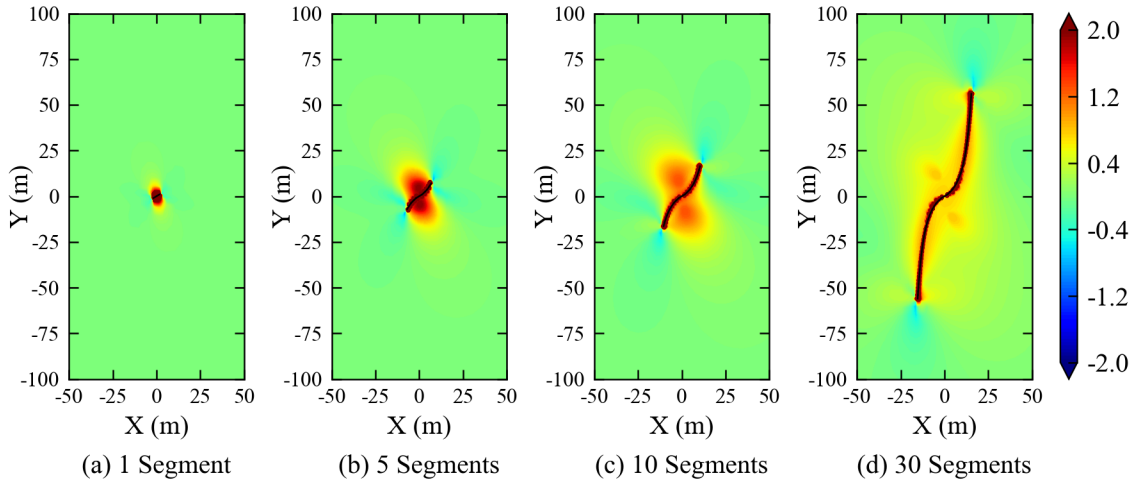


Figure 9. Induced maximum principal stress fields ($\sigma_{max} - \sigma_H$) at various fracture segments (1, 5, 10, and 30) during the propagation process. The simulation is performed with $\theta = 30^\circ$ and $q = 3 \text{ m}^3 / \text{min}$, where black solid lines denote the fracture propagation path.

Source: Authors own work

5 Conclusion

In this paper, we propose a hybrid framework integrating PINNs with the DDM. This approach effectively mitigates the convergence difficulties arising from the high sensitivity of NR solvers to initial values in hydraulic fracture propagation simulations. By embedding physical constraints into the loss function of the neural network, the

model is trained to capture the underlying mechanisms of fracture propagation, thereby predicting high-quality initial values. Experimental results demonstrate that the MSE of physical variables consistently converges to the order of 10^{-6} . Moreover, compared with random initialization, the proposed model eliminates the 27.22% divergence rate and reduces the average number of solver iterations by a factor of approximately 8, thus significantly enhancing computational efficiency.

This study validates the effectiveness of integrating PINNs with traditional numerical methods to solve strongly nonlinearity and multi-physics coupled fracture problems, serving as a transferable paradigm for mitigating initial value sensitivity in other complex coupled systems. Currently, the implementation focuses on single symmetric fracture propagation in two dimensions. Future work will extend this framework to multi-fracture competitive propagation and complex three-dimensional scenarios, providing a powerful tool for optimizing hydraulic fracturing designs in practical engineering.

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Appendix A Governing Equations for Rock Deformation

According to the plane strain theory of elasticity, the stress components induced by shear displacement discontinuity D_s and normal displacement discontinuity D_n for a

fracture segment in its local coordinate system (x, y) are expressed as (Crouch Steven L et al. 1983):

$$\begin{cases} \sigma_{xx} = 2GD_s \left[2 \frac{\partial^2 f}{\partial x \partial y} + y \frac{\partial^3 f}{\partial x \partial y^2} \right] + 2GD_n \left[\frac{\partial^2 f}{\partial y^2} + y \frac{\partial^3 f}{\partial y^3} \right], \\ \sigma_{yy} = 2GD_s \left[-y \frac{\partial^3 f}{\partial x \partial y^2} \right] + 2GD_n \left[\frac{\partial^2 f}{\partial y^2} - y \frac{\partial^3 f}{\partial y^3} \right], \\ \sigma_{xy} = 2GD_s \left[\frac{\partial^2 f}{\partial y^2} + y \frac{\partial^3 f}{\partial y^3} \right] + 2GD_n \left[-y \frac{\partial^3 f}{\partial x \partial y^2} \right], \end{cases} \quad (\text{A-1})$$

where G is the shear modulus, and the function $f(x, y)$ is defined as:

$$\begin{aligned} f(x, y) = \frac{-1}{4\pi(1-\nu)} & \left[y \left(\arctan \frac{y}{x-a} - \arctan \frac{y}{x+a} \right), \right. \\ & \left. -(x-a) \ln \sqrt{(x-a)^2 + y^2} + (x+a) \ln \sqrt{(x+a)^2 + y^2} \right], \end{aligned} \quad (\text{A-2})$$

where ν is Poisson's ratio and a is the half-length of the segment.

From the fundamental solution (A-1), the local induced normal stress σ_n^{ij} and tangential stress σ_s^{ij} produced by segment j at the center of segment i are given by (Crouch Steven L et al. 1983):

$$\begin{cases} \sigma_n^{ij} = 2GD_s^j \left(-y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right) + 2GD_n^j \left(\frac{\partial^2 f}{\partial y^2} - y_{ij} \frac{\partial^3 f}{\partial y^3} \right), \\ \sigma_s^{ij} = 2GD_s^j \left(\frac{\partial^2 f}{\partial y^2} + y_{ij} \frac{\partial^3 f}{\partial y^3} \right) + 2GD_n^j \left(-y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right), \end{cases} \quad (\text{A-3})$$

where (x_{ij}, y_{ij}) denotes the coordinates of the center of segment i expressed in the local coordinate system of segment j .

To aggregate the stress contributions from all segments on the i -th segment, the induced stresses are converted to the global coordinate system according to (Crouch Steven L et al. 1983):

$$\begin{bmatrix} \sigma_{XX} & \sigma_{XY} \\ \sigma_{YX} & \sigma_{YY} \end{bmatrix} = \mathbf{A}^T \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix} \mathbf{A}, \quad (\text{A-4})$$

where the transformation matrix $\mathbf{A}$ is defined as:

$$\mathbf{A} = \begin{bmatrix} \cos \theta_{x_j-X} & \sin \theta_{x_j-X} \\ -\sin \theta_{x_j-X} & \cos \theta_{x_j-X} \end{bmatrix},$$

and θ_{x_j-X} denotes the angle between the local x -axis of the j -th segment and the global X -axis.

By inserting Eq. (A-3) into Eq. (A-4), we obtain the induced stress expressions exerted by segment j on segment i in the global coordinate system (Olson 2004):

$$\begin{cases} \sigma_n^{ij} = 2GC_{ns}^{ij}D_s^j + 2GC_{nn}^{ij}D_n^j, \\ \sigma_s^{ij} = 2GC_{ss}^{ij}D_s^j + 2GC_{sn}^{ij}D_n^j, \end{cases} \quad (\text{A-5})$$

where the influence coefficients are defined as:

$$\begin{cases} C_{ns}^{ij} = \sin^2 \theta_{ij} \left(2 \frac{\partial^2 f}{\partial x \partial y} + y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right) - \frac{1}{2} \sin 2\theta_{ij} \left(\frac{\partial^2 f}{\partial y^2} + y_{ij} \frac{\partial^3 f}{\partial y^3} \right) + \cos^2 \theta_{ij} \left(-y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right), \\ C_{nn}^{ij} = \sin^2 \theta_{ij} \left(\frac{\partial^2 f}{\partial y^2} + y_{ij} \frac{\partial^3 f}{\partial y^3} \right) - \frac{1}{2} \sin 2\theta_{ij} \left(-y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right) + \cos^2 \theta_{ij} \left(\frac{\partial^2 f}{\partial y^2} - y_{ij} \frac{\partial^3 f}{\partial y^3} \right), \\ C_{ss}^{ij} = -\frac{1}{2} \sin 2\theta_{ij} \left(2 \frac{\partial^2 f}{\partial x \partial y} + 2y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right) + \cos 2\theta_{ij} \left(\frac{\partial^2 f}{\partial y^2} + y_{ij} \frac{\partial^3 f}{\partial y^3} \right), \\ C_{sn}^{ij} = -\frac{1}{2} \sin 2\theta_{ij} \left(2y_{ij} \frac{\partial^3 f}{\partial y^3} \right) + \cos 2\theta_{ij} \left(-y_{ij} \frac{\partial^3 f}{\partial x \partial y^2} \right). \end{cases}$$

Furthermore, to incorporate 3D effects into the stress field calculations, a correction factor dependent on the fracture height h is applied (Olson 2004):

$$G^{ij} = 1 - \frac{d_{ij}^\beta}{\left[d_{ij}^2 + (h / \lambda)^2 \right]^{\frac{\xi}{2}}}. \quad (\text{A-6})$$

During fracture propagation, stress equilibrium is satisfied on the surface of each fracture segment. Combined with the symmetry, the system of linear equations is given by:

$$\begin{cases} \sum_{j=1}^N (K_{ns}^{ij} D_s^j + K_{nn}^{ij} D_n^j) = p_i - \sigma_h \cos^2 \theta_i - \sigma_H \sin^2 \theta_i, \\ \sum_{j=1}^N (K_{ss}^{ij} D_s^j + K_{sn}^{ij} D_n^j) = \sigma_s^{\text{local}} - \frac{1}{2} (\sigma_h - \sigma_H) \sin 2\theta_i, \end{cases} \quad (i = 1, \dots, N) \quad (\text{A-7})$$

where p_i is the fluid pressure of the i -th fracture segment, σ_h and σ_H are the minimum and maximum horizontal principal stresses, respectively, θ_i is the inclination angle of the i -th fracture segment relative to the x -axis, and the influence coefficients $K_{\alpha\beta}^{ij}$ are defined in Eq. (1).

Appendix B Discretization of Equations (3) and (4)

Since the fracture propagation governing equations described in Section 3 constitute a system of differential-algebraic equations, to facilitate coupling, we employ the Finite Difference Method (FDM) to unify the continuity equations (3) and (4) into algebraic equations.

First, the fluid flow equation (3) can be rewritten in a simplified form by substituting the material parameters:

$$\frac{\partial p}{\partial s} = \frac{12\mu q}{\beta_1 \beta_2 h w^3}. \quad (\text{B-1})$$

According to the first-order forward difference scheme, the discretization of the pressure gradient is given by:

$$\frac{p_w - p_1}{\Delta x} = \frac{12\mu q_0}{\beta_1 \beta_2 h w_0^3}, \quad (\text{B-2})$$

where p_w is the wellbore pressure, μ is the fluid viscosity, q_0 is the injection rate, w_0 is the fracture width at the wellbore, and Δx is the half-length of the fracture segment.

Next, the mass conservation equation (4) comprises flow rate gradient, fluid leak-off, and fracture volume change terms. We begin by discretizing the flow rate gradient $\partial q / \partial s$ using the FDM spatially and the Crank-Nicolson scheme temporally, yielding:

$$\int_{t^{m-1}}^{t^m} \frac{\partial q(s, t)}{\partial s} dt \approx \int_{t^{m-1}}^{t^m} \frac{q_{j-1}(t) - q_j(t)}{\Delta s} dt = \frac{\Delta t}{2\Delta x} \left[(q_{j-1}^m - q_j^m) + (q_{j-1}^{m-1} - q_j^{m-1}) \right], \quad (\text{B-3})$$

where Δt is the time step, and q_j^m is the flow rate of the j -th segment at the m -th time step.

For the fluid leak-off term $\frac{2hC_L}{\sqrt{t - \tau(s)}}$, integrating over the time interval yields:

$$\int_{t^{m-1}}^{t^m} \frac{2hC_L}{\sqrt{t - \tau(j)}} dt = 4hC_L \left(\sqrt{t^m - \tau(j)} - \sqrt{t^{m-1} - \tau(j)} \right), \quad (\text{B-4})$$

where C_L is the leak-off coefficient, t^m represents the time at the m -th time step, and $\tau(j)$ denotes the time at which the j -th fracture segment is generated.

For the fracture volume change term $\partial A / \partial t$, the temporal integration results in:

$$\int_{t^{m-1}}^{t^m} \frac{2hC_L}{\sqrt{t - \tau(j)}} dt = 4hC_L \left(\sqrt{t^m - \tau(j)} - \sqrt{t^{m-1} - \tau(j)} \right), \quad (\text{B-5})$$

where A_j^m denotes the cross-sectional area of the j -th segment at the m -th time step, and $D_n^m(j)$ is the associated normal displacement discontinuity.

Combining these terms, we obtain the discretized mass conservation equation for segment j :

$$\begin{aligned} \frac{\Delta t}{2\Delta x} \left[(q_j^m - q_{j-1}^m) + (q_j^{m-1} - q_{j-1}^{m-1}) \right] + 4hC_L \left(\sqrt{t^m - \tau(j)} - \sqrt{t^{m-1} - \tau(j)} \right) \\ + h \left(D_n^m(j) - D_n^{m-1}(j) \right) = 0. \end{aligned} \quad (\text{B-6})$$

For the initial fracture segment, substituting the conditions $q_1^1 = q_1^0$, $t^0 = \tau(1) = 0$, and $D_n^0(1) = 0$ yields:

$$\Delta t(0 - q_0) + 4hC_L \cdot (2\Delta x)\sqrt{\Delta t} + (2\Delta x)hD_n^1(1) = 0. \quad (\text{B-7})$$

Appendix C Training Loss Curve

Figure 10 depicts the loss curve under the hybrid optimization strategy. The training process begins with the Adam optimizer, resulting in a rapid decay in the early stages that effectively guides the network parameters toward the vicinity of the optimal solution. At epoch 2,000, the transition to the L-BFGS optimizer triggers a precipitous drop in the loss value, breaking through the local stagnation to drive the residuals down to a stable magnitude of $O(10^{-5})$. The convergence behavior confirms that the proposed framework can strictly satisfy the multi-physics constraints with high numerical precision.

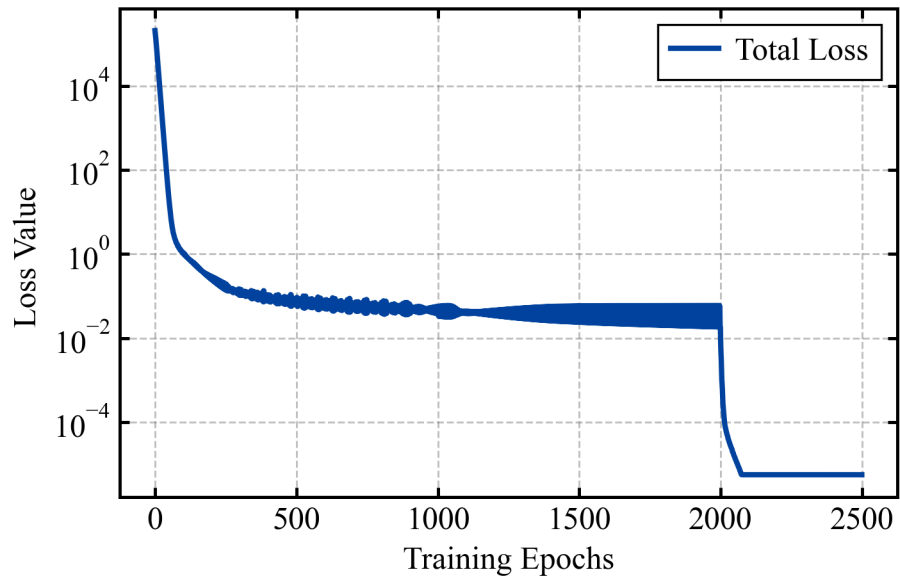


Figure 10. Convergence curve of the total loss function.

Source: Authors own work